LALUAN · TEAM MADE IN UTM · SDC 2026 HACKATHON, DOMAIN 1

Where Laluan's data comes from

Every number and list in the app, with its source, how it was obtained and where it is kept in the repository. As of 20 October 2026. Live app: laluan-psi.vercel.app · Code: github.com/tanhongsheng050204-code/XMUM-HACKATHON

1. Official published sources
Job forecasts (WEF), Malaysian pay (DOSM), tasks (O*NET), SDGs (UN), course prices (providers).

2. Collected live by the app
Research activity (arXiv / OpenAlex) and job-post demand (Hacker News), only after a person approves.

3. Written by the project
Role and skill lists, AI-exposure estimates, task labels, SDG reasons, and the synthetic demo people.

1. Official published sources REAL DATA

Data	Source	How we obtained it	In the repository
Net job growth 2025–2030 for each role (e.g. Data Entry Clerks −25.9%)	World Economic Forum, *Future of Jobs Report 2025*, Figure 2.3 (p. 21). reports.weforum.org/docs/WEF_Future_of_Jobs_Report_2025.pdf	The chart prints no numbers. `scripts/extract_wef_figure.py` reads each marker's position from the official PDF against the chart's gridlines (±0.5 points). Cross-checked with figures quoted in the press: data entry −26%, admin assistants −20%, bank tellers −31%, postal clerks −34% — all matched.	`backend/data/raw/wef_foj2025_figure_2_3.csv` (126 jobs); `backend/data/signals/2026-10-20-wef-foj-2025.json`
Malaysian median pay and employee numbers by occupation group (e.g. clerical RM 2,693, +11.1% since 2019)	Department of Statistics Malaysia (DOSM), *Annual Salaries & Wages by Occupation & Sex*, OpenDOSM, CC BY 4.0. open.dosm.gov.my/data-catalogue/wages_occupation	Official CSV downloaded by `scripts/refresh_malaysia.py`; compared with 2019, the last year before the pandemic.	`backend/data/raw/dosm/`; `backend/data/malaysia_pay.json`
Job tasks (71 task statements)	O*NET 30.0, U.S. Department of Labor / National Center for O*NET Development, CC BY 4.0. onetcenter.org	Task text copied word for word by `scripts/build_tasks.py`.	`backend/data/tasks.json`
SDG goals, targets and colours	United Nations. sdgs.un.org/goals	Names, target wording and official colours.	`backend/data/sdgs.json`
Course prices and HRD Corp status	Providers' own pages: SEDA Malaysia GCPV course (RM 5,660.40 excl. SST) seda.gov.my; PwC Malaysia GRI Standards training brochure, May 2025 (RM 8,000, HRD Corp claimable with employer approval); GHG Protocol (Corporate Standard webinar free) ghgprotocol.org/e-learning	Read from those pages on 20 Oct 2026; each course stores the date checked. Providers without a published fee (NIOSH, electrical, caregiver) show “ask the provider”.	`backend/data/courses.json` (`price_rm`, `hrd_corp`, `price_checked`)

Course names and links (29 courses)	Coursera, Microsoft Learn, Harvard CS50, Khan Academy, freeCodeCamp, AWS Skill Builder, DeepLearning.AI, Elements of AI, The Odin Project, NIOSH, SEDA, GRI, GHG Protocol	Each link opened and dated by `scripts/check_links.py` (27 of 29 have a single course page).	`backend/data/courses.json` (`checked`)
Malaysian context figures (56% use AI at work, 26% trained, 600,000+ need reskilling)	SDC Hackathon *Domain 1 Problem Statement*, citing the Hays Asia Salary Guide 2025 and MOSTI	Quoted from the brief.	`docs/01-PRD.md`, proposal

2. Collected live by the app REAL, UPDATABLE

Pulled when an admin presses **Refresh trend data**. Through official public APIs, not scraping. A new forecast is saved for review and goes live only after a person approves it.

Data	Source	How it is used	In the repository
AI research activity per capability	arXiv API export.arxiv.org/api; OpenAlex as a backup openalex.org (CC0)	Paper counts for the last 6 months vs. a year earlier → which AI capabilities are accelerating → adjusts each role's outlook within a capped range.	`backend/data/raw/research/`
Skill demand in job postings	Hacker News "Ask HN: Who is hiring?" threads via the Algolia API hn.algolia.com/api	Share of postings mentioning each skill, this year vs. last year. Counts only; no posts or personal data stored. Tech-heavy, so non-tech skills show "no data".	`backend/data/raw/jobs/`

3. Written by the project TEAM JUDGEMENT

First drafted with an AI coding assistant (Claude) while building the app, under the team's direction, as declared in `docs/AI_DECLARATION.md`. Labelled as estimates in the app.

Data	How it was made	Checked?	In the repository
13 roles and the skills each one needs (with weights)	Drafted, then compared with O*NET occupations	Partly: O*NET code shown on each role	`backend/data/roles.json`
42 skills and the words used to recognise them in CVs	Drafted	To be measured by the human CV evaluation (target ≥ 80% recall)	`backend/data/skills.json`
AI-exposure scores per role; scenario speeds (Slow 0.5×, Accelerated 1.6×)	Modelling assumptions. Baseline scenario uses WEF figures exactly; Slow and Accelerated are our modelling.	No published source; stated on every role	`backend/data/signals/…wef….json`, `backend/data/scenarios.json`
Automate / assist / human labels on the 71 tasks	Drafted with AI help	Not yet reviewed (0 of 71)	`backend/data/tasks.json`

Which SDG each role and skill serves, with reasons	Drafted	Not yet reviewed	`backend/data/sdg_map.json`
Matching roles to WEF and DOSM categories (e.g. AI operations → WEF "Process Automation Specialists")	Team choice	Recorded with reasons in `docs/DECISIONS.md`	`backend/data/raw/wef_foj2025_roles.csv`, `roles.json`
Course hours	Estimates	Not all checked against providers	`backend/data/courses.json`
Demo people (Aisyah, Wei Jie), 5 evaluation CVs, 160-person sample company	Written as synthetic examples	Synthetic on purpose: the rules forbid real CVs and personal data	`backend/data/demo_profiles.json`, `backend/data/eval/`, `backend/data/sample_workforce.csv`
Original "illustrative seed" forecast	Placeholder numbers used before the WEF data was added	No longer used; kept for history	`backend/data/signals/2026-10-19-illustrative-seed.json`

> **Also from outside the project:** three landing-page photos from Unsplash (Gema Saputera, Mimi Thian, Vitaly Gariev; Unsplash License; the people shown are not the demo personas). Open-source libraries and their licences are listed in `docs/submission/SDC2026_MadeInUTM_TechDeclaration.md`.

Laluan supports career decisions; it does not make them. Forecasts are estimates under different scenarios, not predictions. Every role, task, course and forecast in the app shows its source and date; the Sources page lists every forecast snapshot and who approved it.